

CHARON WHITEPAPER FIGURES

Technical figure draft for runtime boundary enforcement

Figure 1. Agent Action Boundary

Charon evaluates a requested action before the operating system, tool, file, network target, or credential is touched.

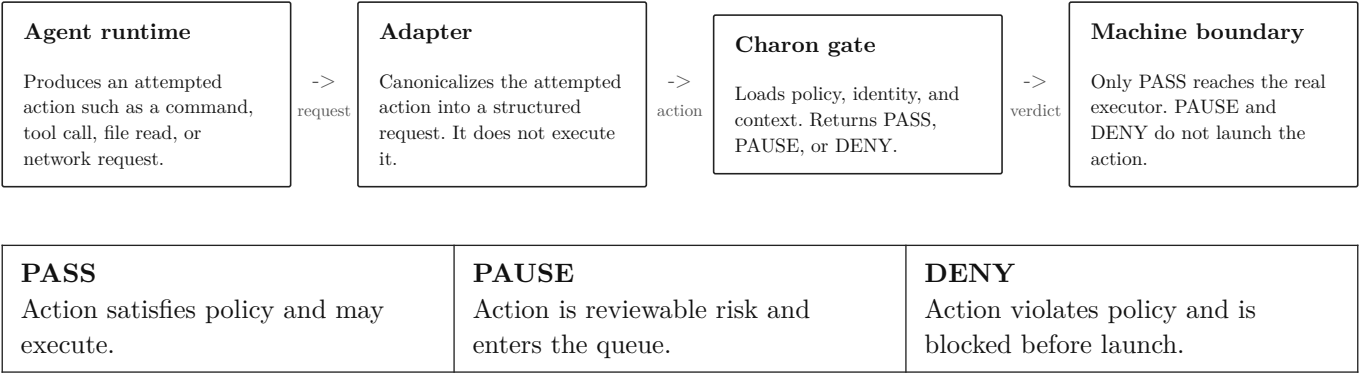


Figure 2. Agent Threat Surface

The critical risk is not only model output. It is the path from untrusted context into external side effects.

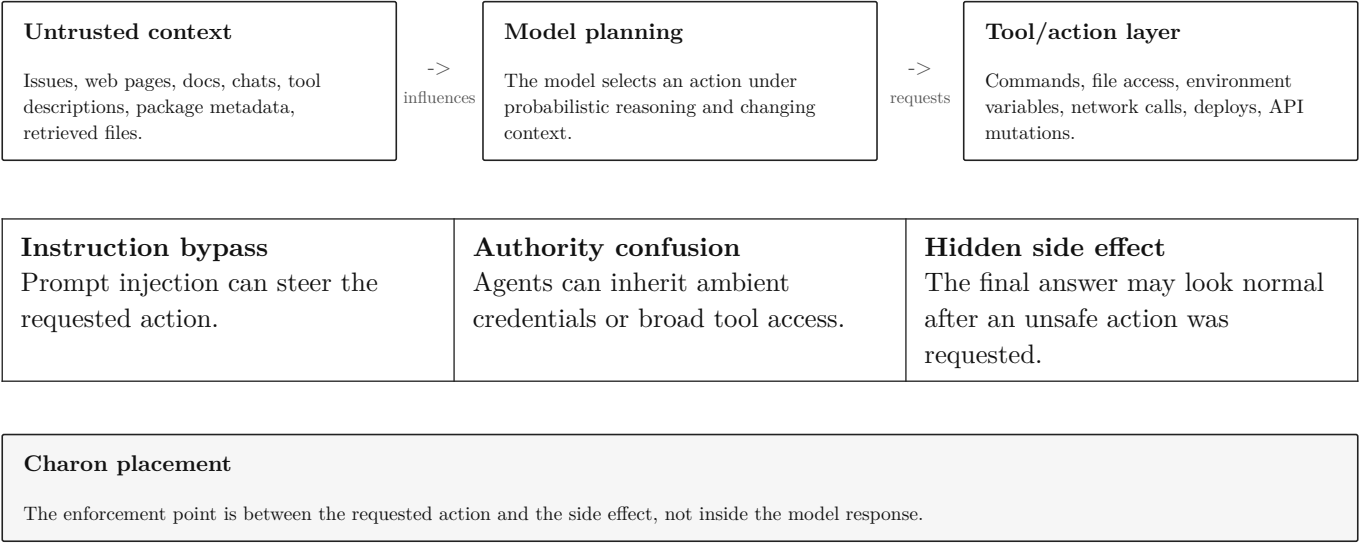


Figure 3. Charon Decision Flow

A requested action is normalized, evaluated, and either executed, queued, or refused. Every branch writes a receipt.

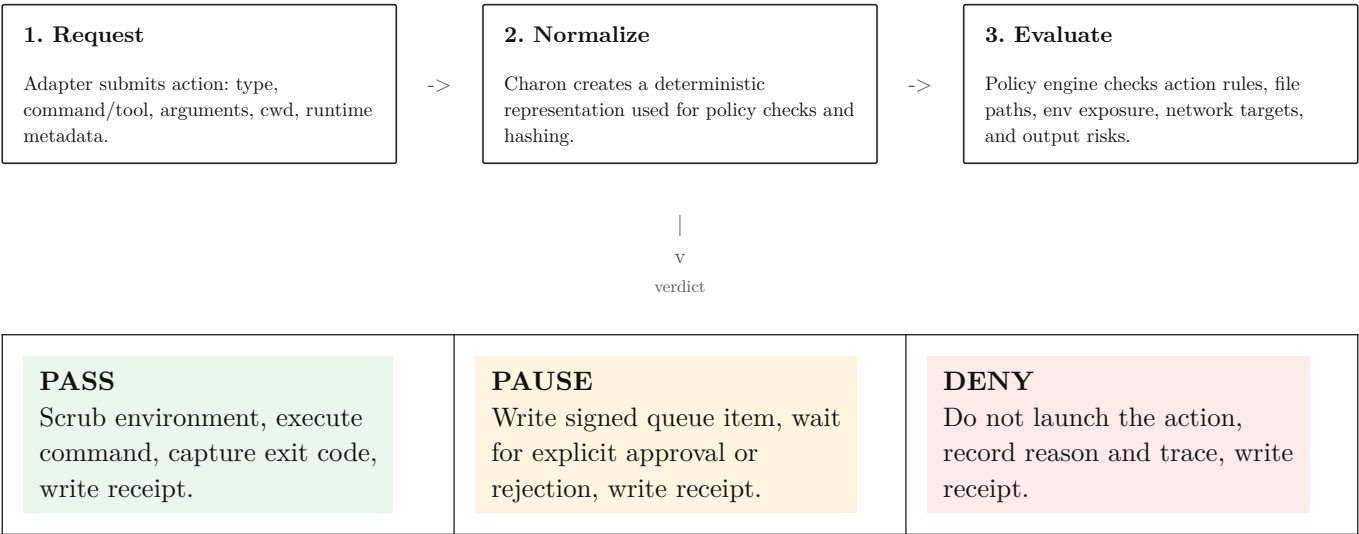


Figure 4. Policy Surface

Charon policy is not a prompt. It is structured control over the action surfaces that produce machine effects.

Surface	What is checked	Example decision
Command	Executable plus arguments after canonicalization.	<code>npm publish</code> -> DENY
File	Requested paths inferred from command and structured tool input.	<code>read:.env</code> -> DENY
Environment	Names exposed to the child process; secret values are never stored.	<code>GITHUB_TOKEN</code> -> withheld
Network	Hosts or URLs requested by action text or tool input.	<code>webhook.site</code> -> DENY
Release action	Operations that mutate public state or production state.	<code>git push</code> -> PAUSE
Output	Secret-like output is redacted and can convert PASS into DENY.	<code>github_pat_...</code> -> redacted

Figure 5. Receipt Anatomy

A receipt is the durable evidence for one Charon decision. It binds action, verdict, policy, identity, and trace.

Action fields command, cwd, runtime metadata, adapter metadata	Decision fields verdict, reason, exit code, startedAt, endedAt
Policy binding policyHash over normalized policy controls and bounds	Boundary trace matched action rule, file result, network result, secret result
Redaction record commandRedactions, reasonRedactions, output redactions	Proof fields receipt signature, identity action hash, identity signature

Important constraint
Receipts should prove what Charon saw and decided. They should not store secret values.

Figure 6. Signed Receipt Verification

Verification recomputes hashes and checks signatures. Tampering with the receipt body or identity-bound action breaks verification.

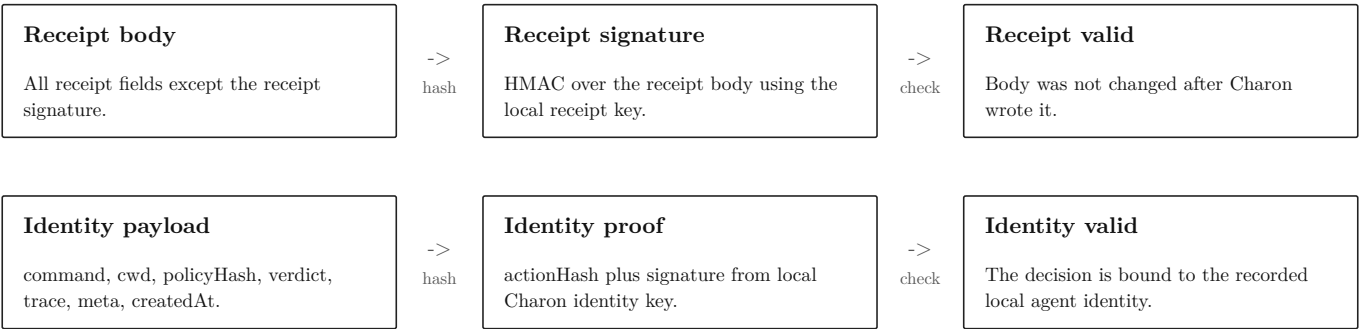


Figure 7. Before And After Charon

Charon changes the execution shape from direct tool access to mediated action access with explicit evidence.

Without Charon Agent runtime -> tools -> machine side effects Controls are usually prompts, framework allowlists, broad credentials, and post-hoc logs.	With Charon Agent runtime -> adapter -> Charon gate -> PASS / PAUSE / DENY -> receipt Controls are evaluated before launch, risky actions queue, denied actions do not execute, and decisions are verifiable.
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Before launch policy evaluation	At decision queue or execution	After decision signed receipt
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